J Laxmi Bhavani Saraswathi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.12
Sri Chaitanya Jr Kalasala	Class XII (TGBIE)	2024	98.20%
Genesis International School	Class X (CBSE)	2022	97.00%

POSITIONS OF RESPONSIBILITY

UG - Buddy, IITH Sunshine Club

SKILLS

EDA Tools - Cadence Virtuoso, LTspice

PCB Design - Altium Designer, KiCAD

Digital Design - Verilog

Programming Languages - C

Embedded Systems - Arduino

RELEVANT COURSES

Lab Courses: Digital System Lab, Electrical Circuits lab, Semiconductor Device Lab, Electronic System Design Project Lab,

PCB Design and Layout for electronic systems, Analog Lab

Computational and Mathematics: Data Structures & Applications, Introduction to Programming, Vector Calculus, Probability and Random Processes, Matrix Theory, Complex Variables.

Electronics: Circuits and Network Analysis, Digital Systems, Semiconductor Device Fundamentals, Analog Electronics,

Numerical Methods for Device Modelling, Productivity Tools for IC Design, Engineering Electromagnetics,

Signals and Systems, Digital Fabrication,

Probability and Random processes, Machine Learning, Introduction to CMOS Processing.

PROJECTS

Characterization of CMOS and NMOS Logic Gates

EE2501: Semiconductor Devices Lab

- Hardware implementation and characterization of universal gates NAND and NOR using nMOSFETs and pMOSFETs and plotted transfer characteristics for inverter circuit using PWM from Arduino to capture the data and plotting it.

- Also measured Noise Margin and found CMOS-based gates had better Noise Margin and Low Power Dissipation.

PCB Design of Analog RF Signal Generator

EE4130: PCB Design and Layout for Electronic Systems

- Designed and implemented a PCB for an analog RF signal generator operating up to 5 MHz.

- Integrated Colpitts oscillator, RC phase shift oscillator, and modulation stage for signal generation and processing.

Digital Systems Design | Verilog HDL

EE1501 Digital System Lab

Traffic Signal Controller

- Designed a T-junction traffic signal controller using FSMs in Verilog.

- Implemented pedestrian walk signals and emergency vehicle priority for traffic control.

- Verified traffic light transitions and emergency scenarios using a Verilog testbench and timing diagrams.

Digital Clock System

- Designed a digital clock in Verilog supporting 12-hour and 24-hour modes with AM/PM indication.

- Implemented a countdown timer, programmable alarm, and date display using sequential logic.

- Developed the design using finite state machines and verified functionality through simulation and timing waveforms.

Analog Circuit Design and Simulation using LTspice

EE2400 - Analog Electronics

- Designed and simulated **Schmitt trigger, Pierce oscillator, LDO, and multivibrator circuits** to study fundamental analog and mixed-signal circuit behavior.

- Analyzed hysteresis, switching thresholds, feedback mechanisms, oscillation frequency, and transient responses through time-domain simulations.

- Evaluated simulated waveforms and circuit characteristics to understand the practical behavior of analog building blocks.

5-bit R-2R DAC and SAR ADC

EE2400 Analog Electronics

- Designed and implemented a 5-bit R-2R ladder DAC, and characterized DAC operation by applying different binary input combinations and measuring the resulting analog output voltage.

- Designed and implemented a 5-bit Successive Approximation Register (SAR) ADC using discrete components on a breadboard.

- Verified the DAC operation experimentally by comparing measured conversion levels with theoretical values on the CRO.

EXTRACURRICULAR

Member, National Sports Organization (NSO) - Badminton